The Shadow Price of Neutrality

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***Abstract*—A multi-manager platform caps each long/short pod's ex-ante volatility (3%/yr), imposes a neutrality mandate, and reads each rule's cost from its dual, exact for convex programs but not, in general, with a cardinality limit and minimum positions. This single-period study prices each rule of an S&P 500 momentum book at the fixed risk budget, scores three cheap marginal-cost estimators under those limits against solver-certified secant intervals, and compares big-M and perspective formulations. Most of the cost comes from the liquidity cap (median elasticity 0.16) and from trading and borrow; factor neutrality is cheap because the alpha is weakly aligned with the factors, so idealized full factor neutrality keeps at least 0.964 of the unconstrained, costless optimum. The dual with the holdings fixed is exact if no holding or active constraint changes over the step (maximum error 6.7×10^{-12}, 7 cases), and the best estimator depended on the step size at one date. At default scaling, the perspective model's root bound is consistent with Gurobi's outer approximation stopping early (forced, it reproduces the bound at 29 of 36 instances) and the cone tolerance inflates its incumbents; rescaling rows and cones together removes both, and presolve, not the formulation, closes big-M's root gap.**

I. Introduction

A multi-manager platform allocates capital to long/short equity teams (pods). To keep one pod's risk off the others' books, it caps each pod's volatility (3%/yr here) and imposes a neutrality mandate on its dollar, beta, style and sector exposures. A binding rule costs the pod expected alpha, which the platform reads from the rule's dual variable; in a convex program the reading is exact where the dual is unique. Limits on how many names the book holds and how small a held position may be make the program mixed-integer, and a continuous dual can then misstate a rule's marginal cost (Sec. IV-B).

Clarke, de Silva and Thorley measure with a transfer coefficient how constraints dilute an alpha forecast [1], building on Grinold and Kahn's framework [2], and Stubbs and Vandenbussche attribute a portfolio to its constraints through their duals [3]. Misaligned risk and alpha factors distort optimized portfolios [4], [5]. Related models treat realistic shorts [6] and sparse portfolios [7]. Perspective cuts and reformulations strengthen mixed-integer programs with indicator variables, including portfolios with position thresholds [8], [9]. Geoffrion and Nauss survey postoptimality analysis for integer linear programs [10], and Wolsey shows that integer duality needs price functions rather than prices [11]; standard texts give the background [12], [13].

This paper prices each constraint by its own dual at a fixed risk budget, so costs compare at equal risk (in the integer problem, at the fixed risk price λ^*; Sec. II-E), and asks which constraints bind and what they cost (Q1), how well continuous tools predict marginal costs under integrality (Q2), and how the formulations and solver tolerances interact (Q3; Sec. II-G). It contributes: (i) costs at equal risk, with the closed form $\sqrt{1-\varphi}$ for the alpha kept under full factor neutrality (φ: the factors' share of the alpha's squared information ratio); under idealized neutrality the momentum alpha keeps at least 0.964 ($n = 12$ dates), and at the base mandate the cost comes mainly from the liquidity cap (median elasticity 0.16) and the trading and borrow costs; (ii) secant intervals certified by the solver's bounds (median relative width 3.6×10^{-4}, $n = 23$) that separate estimator error from solver error; (iii) when the fixed-integer dual (the dual with the incumbent's holdings fixed) is exact: error at most 6.7×10^{-12} ($n = 7$) when neither the names held nor the active set changes within the step, median 0.056 ($n = 24$) when the holdings change; (iv) two default-scaling solver effects in the perspective model, kept apart: a root bound consistent with Gurobi choosing a polyhedral outer approximation of the cones (29 of 36 instances) that stops early (median root gap 0.015, against 5.7×10^{-4} with QCP node relaxations), and incumbents inflated by the cone tolerance (at default presolve, up to 6.4×10^{-3} of the objective, against 1.9×10^{-11} with rows and cones rescaled together). The study solves a single-period problem at each date on one universe and risk model, with no backtest or performance claim.

II. Problem Setting

A. Decision problem

A *pod* receives each month a forecast α_i of each stock's (*name*'s) *alpha*, its expected return beyond what its factor exposures explain, and chooses weights w as fractions of AUM (assets under management, \$1B). The platform caps the *ex-ante volatility*, the risk model's forecast $\sqrt{w^\top \Sigma w}$ annualized, at 3%/yr ($\sigma_m = 3\%/\sqrt{12}$ per month) and imposes a mandate. The *factor exposure* is $y = B^\top w$, B the loadings on the market and five styles (size SMB, value HML, profitability RMW, investment CMA, momentum Mom); *beta* is y_{mkt}. *Dollar*, *beta*, style and *sector neutrality* bound the net weight $|1^\top w|$, $|y_{\mathrm{mkt}}|$, each $|y_k|$ and each $|S_j^\top w|$, S_j the 0/1 indicator of GICS sector j (11 sectors). *Gross leverage* $1^\top(w^+ + w^-)$ is the long plus the short weight. A stock's *ADV* is its median daily dollar volume over 63 trading days; with the position cap u, the *liquidity cap* (a fraction θ of ADV) gives $u_i = \min(u, \theta\,\mathrm{ADV}_i/\mathrm{AUM})$. Shorts pay a *borrow cost* c per month (per year in Table I) and trades a one-way *trading cost* κ (1 bp $= 0.01\%$); from cash ($w_0 = 0$) trading costs κ times gross. Table I lists the base mandate.

B. Data and risk model

Daily adjusted prices and volumes of the current S&P 500 constituents (500 tickers, 2005-01-04 to 2025-12-31) come from yfinance, sectors from the constituent list, and the daily Fama–French five factors, momentum and risk-free rate from the French library [14]–[16]. A day is valid for a stock with a return and a dollar volume of at least \$1M; a stock is *eligible* at a date if at least 80% of the 252 trading days ending there are valid. At each of the 241 month-ends (2005-12 to 2025-12) eligible stocks' daily *excess returns* (minus the risk-free rate) over that window are regressed on the six factors: B holds the loadings, $D = \mathrm{diag}(d)$ the residual variances and F the sample factor covariance, both scaled by 21 trading days to monthly, with d floored at its 5th cross-sectional percentile, so $\Sigma = BFB^\top + D \succ 0$. The eligible universe has 373–499 names.

C. Alphas and alignment

The momentum score, each stock's return over the 252 trading days before the date skipping the last 21 (it exists at 240 of the 241 month-ends), is z-scored across eligible stocks, winsorized at ± 3 and re-standardized; the alpha is $\alpha_i = \mathrm{IC} \cdot \omega_i z_i$ with information coefficient $\mathrm{IC} = 0.03$ and residual volatility $\omega_i = \sqrt{d_i}$ [2]. A D^{-1}-weighted regression of an alpha α on the loadings splits it into a residual $\alpha_\perp = \alpha - B(B^\top D^{-1} B)^{-1} B^\top D^{-1} \alpha$ and a factor-spanned part $\alpha_\parallel = \alpha - \alpha_\perp$. The *alignment share* $\varphi(\alpha) = \alpha_\parallel^\top \Sigma^{-1} \alpha_\parallel / \alpha^\top \Sigma^{-1} \alpha \in [0,1]$, on α's own, unrescaled split, is the factors' part of $\alpha^\top \Sigma^{-1} \alpha$, the squared *information ratio* (the best ratio of expected alpha to ex-ante risk); rescaling α leaves it unchanged. α_{mom}'s $\alpha_\perp$ and $\alpha_\parallel$, each rescaled to α_{mom}'s information ratio, also serve as signals. The *transfer coefficient* $\mathrm{TC} = \alpha^\top w / (\sqrt{\alpha^\top \Sigma^{-1} \alpha} \sqrt{w^\top \Sigma w})$ is the Σ-metric form of Clarke et al.'s [1].

D. The convex problem P-RB

The risk-budget problem P-RB, with $w = w^+ - w^-$, $w^\pm \geq 0$, is

$$\begin{aligned} \max\ & \alpha^\top w - c\, 1^\top w^- - \kappa\, 1^\top (w^+ + w^-) \\ \text{s.t. } & y^\top F y + \textstyle\sum_i d_i w_i^2 \leq \sigma_m^2, \quad y = B^\top w, \\ & |1^\top w| \leq \varepsilon_{\mathrm{net}},\ |y_{\mathrm{mkt}}| \leq \varepsilon_\beta,\ |y_k| \leq b_k,\ |S_j^\top w| \leq h_j, \\ & 1^\top (w^+ + w^-) \leq G, \quad w_i^+, w_i^- \leq u_i. \end{aligned} \tag{1}$$

Since $c, \kappa > 0$, at most one of $w_i^\pm$ is positive at an optimum. The optimal value A^* is net expected alpha per month. Row k has multiplier μ_k; a parameter p (a bound, u or θ) has the *family dual* $\mu_p = dA^*/dp = \sum_k \mu_k \, \partial r_k / \partial p$ over the rows whose right-hand side r_k moves with p (A5): θ through every θ-limited cap ($\partial u_i / \partial \theta = \mathrm{ADV}_i / \mathrm{AUM}$), u through every cap equal to u, a common sector or all-styles bound through all its rows. A *key* is a parameter whose dual is reported: a family, or one sector's or style's row. The *elasticity* $e_p = \mu_p p / A^*$ ($A^* > 0$) is the percent change in alpha per percent change in p; a constraint *binds* if $e_p \geq 10^{-4}$. The risk row's multiplier λ^* is the price of risk at which a mean-variance investor picks the same portfolio (A1). A fixed risk budget, not a fixed λ, compares costs at equal risk: relaxing a constraint cannot buy alpha by levering up.

E. The integer problem P-MI

The mixed-integer problem P-MI adds a cardinality limit K and a minimum position $\ell = 0.25\%$, moves the risk row into the objective at λ^*, and raises the position cap to $u = 6\%$:

$$\begin{aligned} \max\ & \alpha^\top w - \lambda^* \big(y^\top F y + \textstyle\sum_i d_i w_i^2 \big) \\ & \quad - c\, 1^\top w^- - \kappa\, 1^\top (w^+ + w^-) \\ \text{s.t. } & \text{the mandate of (1) without the risk row,} \\ & \ell z_i^\pm \leq w_i^\pm \leq u_i z_i^\pm,\ z_i^+ + z_i^- \leq 1,\ \textstyle\sum_i (z_i^+ + z_i^-) \leq K, \\ & z^\pm \in \{0,1\}^N. \end{aligned} \tag{2}$$

λ^* is P-RB's multiplier on the same universe at the base mandate with $u = 6\%$, fixed for that date and universe: a Lagrangian relaxation of the risk row, with zero gap in the convex case. *Big-M* keeps $\sum_i d_i w_i^2$ with $M = u_i$; the *perspective* formulation replaces it by $\sum_i d_i q_i$ with rotated cones $w_i^2 \leq q_i \zeta_i$, $q_i \geq 0$, $\zeta_i = z_i^+ + z_i^-$; the *rescaled perspective* divides every weight, row and bound by $\max_i u_i$ and writes each cone in its name's bound, $t_i = w_i / u_i$, $q_i' = q_i / u_i^2$, $t_i^2 \leq q_i' \zeta_i$, which leaves the feasible set and objective unchanged but changes what Gurobi's absolute tolerances act on (A9). These are the three *arms* of N1–N3. With λ^* fixed, P-MI's ex-ante volatility drifts: median 2.87%/yr in I1 (2.42–3.02, $n = 144$, 3 above the budget) and 2.77% in N1 (2.22–3.31, $n = 36$ per arm, 4 above), so Q2's slopes are of the risk-penalized objective. The *continuous baseline*, P-MI with $\ell = 0$ and no cardinality row, is mean-variance at λ^*: with the risk row binding it has P-RB's optimizer and constraint duals at the base point (A1), but the value $A^* - \lambda^* \sigma_m^2$.

F. Problem classes

P-RB is a convex quadratically constrained program (QCQP); its risk row is the second-order cone $\|(L_F^\top y, \sqrt{d} \circ w)\|_2 \leq \sigma_m$ with $F = L_F L_F^\top$. When $\lambda^* > 0$ its optimal w is unique: by A1, w also maximizes the mean-variance objective at $\lambda = \lambda^*$ over the mandate, which is strictly concave in w ($\Sigma \succ 0$, and the costs are convex and piecewise linear in w); the split $w^\pm$ is then unique because $c, \kappa > 0$. The big-M P-MI is a convex mixed-integer QP (every node relaxation a convex QP); the perspective P-MI is a mixed-integer second-order cone program.

G. Questions

Q1 (convex): at a fixed risk budget, which mandate constraints bind, what does each cost, and what determines that cost? **Q2** (integer): with a cardinality limit and a minimum position, how well do relaxation and fixed-integer duals predict a constraint's true marginal cost? **Q3** (formulation and numerics): how do the big-M and perspective formulations compare as the mandate tightens, and how do solver tolerances interact with problem scaling?

TABLE I
BASE MANDATE (P-RB) AND P-MI PARAMETERS; STYLE BOUNDS IN LOADING UNITS.

Parameter	Base
Risk budget (ex-ante volatility)	3%/yr
Gross leverage G	2
Net (dollar) ε_{net}	$\pm$2%
Beta ε_β	$\pm$0.05
Each style b_k	$\pm$0.10
Each sector h_j (net)	$\pm$2%
Position cap u (P-MI)	3% (6%)
Liquidity θ	5% of ADV
AUM	\$1B
Borrow c	30 bp/yr
Trading κ (one-way)	5 bp
Minimum position ℓ (P-MI)	0.25%

III. SOLUTION APPROACH AND ANALYSIS

A. Experiments

Table II defines the experiments: C1–C2 answer Q1, I1–I2 Q2, N1–N3 Q3. D12 is 12 month-ends (2008-10, 2009-03, 2011-08, 2013-06, 2015-08, 2017-12, 2018-12, 2020-03, 2020-11, 2022-06, 2023-03, 2025-04) spanning calm and stressed markets. C1's 72 idealized solves are two mandates $\times$ three alphas $\times$ 12 dates: (i) all six exposures zero, (ii) the same plus dollar neutrality, each with the risk budget only. C2 sweeps one family at a time, the others at base, over (off first): dollar 0.1, 0.05, 0.02, 0.01, 0; beta 0.2, 0.1, 0.05, 0.02, 0; each style and all styles 0.4, 0.2, 0.1, 0.05, 0.02, 0; sector 0.08, 0.04, 0.02, 0.01, 0; gross 4, 3, 2, 1.5, 1; u 0.06, 0.03, 0.015; θ 0.1, 0.05, 0.025. N counts names (top N by ADV); $K = \text{round}(fS)$, half up, S the support size of P-RB at $u = 6\%$. A rule fixed in a pilot run before the experiments chose the 4 I1 bounds: among the families with the largest median C1 elasticity at the I1 dates (u and θ excluded: they set the big-M constants) and the Mom bound at 0.025, those certified binding (A8) at $f = 0.5$ on 2017-12.

B. Algorithms

P-RB is solved as the mean-variance QP $\max \alpha^\top w - \lambda w^\top \Sigma w -$ costs by Gurobi's dual simplex (tolerances 10^{-9}: a basic solution, its active set and duals exact up to them), with λ root-found by Brent's method on $\log \lambda$ until volatility meets the budget to relative 10^{-10} (A1). Clarabel's primal–dual interior-point method [17] (tolerances 10^{-12}) solves the cone form of (1) as a cross-check and the perspective relaxation of (2) ($z \in [0, 1]$), whose finite differences give its slope; dual simplex solves the big-M relaxation and the fixed-integer QP (P-MI with z fixed at the incumbent). *Plain relaxations* are the $z \in [0, 1]$ relaxations solved outside the MIP, without its presolve or root cuts. Gurobi's branch and bound [18] stops when the relative gap is below *MIPGap* (10^{-6} in I1, I2; 10^{-4} in N1, N2(a)) or its *Work*, a deterministic effort count, reaches $W^* = 340$ ($2W^*$ in I1, I2), the Work of 300 s at the pilot's median 1.12 units/s, to two significant figures; MIQCPMethod (cone handling at the nodes) stays at its default. The objective is multiplied by $s_{\text{obj}} = 10^4$, the power of ten that brings the pilot's proxies for $|U^*|$ (the mean-variance value at λ^* on each I1 and N1 universe) to 9.1–21; values are reported unscaled. The incumbent is *polished*: the fixed-integer QP at its z, re-solved at tolerances 10^{-9}, gives V. N2(b) runs Gurobi's barrier and CVXPY's [19] Clarabel and SCS (ECOS is not installed) at default tolerances against Clarabel at 10^{-10}.

TABLE II
EXPERIMENTS; SOLVES: THE OPTIMIZATION SOLVES IN EACH RUN.

exp.	instances	varied, arms	solves
C1	D12 (12 dates); all eligible names; α_{mom}, $\alpha_\perp$, $\alpha_\parallel$	base mandate, and the idealized mandates (i), (ii); envelope at $\delta = 1\%$ (0.1%)	108
C2	D12; all eligible; α_{mom} ($\alpha_\perp$, $\alpha_\parallel$ on factor families)	one family swept over its grid, the others at base	2040
I1	2009-03, 2017-12, 2020-03; $N = 250$; $f \in \{0.25, 0.5, 0.75\}$	4 bounds at $\delta = 10\%$ (also 5% at 2017-12); big-M and rescaled perspective	288
I2	2017-12 ($K = 67$); 2020-03 ($K = 79$); $N = 250$	sector bound at 41 points on $[0, 2b]$; as I1	164
N1	2017-12, 2020-03; $N \in \{100, 250, all\}$; $f \in \{0.25, 0.5\}$; 3 tightness levels	big-M, perspective (default scaling, rescaled); presolve off at $N = 250$; 3 reruns	147
N2(a)	N1 at $N = 250$, base tightness	the N1 arms at $s_{\text{obj}} \in \{1, 10^4\}$	24
N2(b)	C1 base at 5 dates	Gurobi barrier, Clarabel, SCS; risk-row normalization and objective scaling on/off; reference	65
N3	the N1 instances at default presolve	root only, both perspective arms; MIQCPMethod -1, 0, 1 (default scaling also FeasibilityTol 10^{-9})	252

C. Analysis

A1 (risk budget and mean-variance). Dualizing only the risk row, with multiplier $\lambda \geq 0$, gives the partial Lagrangian $\alpha^\top w -$costs$-\lambda(w^\top \Sigma w - \sigma_m^2)$; maximizing it over the mandate is mean-variance at λ. For $\lambda > 0$ its maximizer $w(\lambda)$ is unique ($\Sigma \succ 0$), with volatility continuous, nonincreasing in λ and tending to 0 as $\lambda \to \infty$ ($w = 0$ is feasible), so if the volatility at $\lambda_{\min} = 10^{-6}$ exceeds the budget a root λ^* exists. There the risk row is tight (primal feasibility, complementary slackness), so $w(\lambda^*)$ satisfies the KKT conditions, which suffice for this convex program: it solves P-RB with multiplier λ^*. Slater's condition holds ($w = 0$ strictly satisfies the risk row; the other rows are affine), so strong duality holds and, conversely, every P-RB optimum maximizes the Lagrangian at λ^* (used in Sec. II-F). Otherwise $\lambda^* \leq \lambda_{\min}$ (the budget is treated as slack) and the solver returns $w(\lambda_{\min})$, feasible and with net alpha within $\lambda_{\min}\sigma_m^2 = 7.5 \times 10^{-11}$ of P-RB's optimum (4 of 36 C1 base solves, all $\alpha_\parallel$).

A2–A3 (equality case). With only the rows $Aw = 0$ (A of full row rank), no costs and no other rows, stationarity gives $w^* = \Sigma^{-1} g/(2\lambda)$ with $g = \alpha - A^\top \nu$; $Aw^* = 0$ gives $\nu = (A\Sigma^{-1}A^\top)^{-1} A\Sigma^{-1}\alpha$. If $g \neq 0$ ($\alpha \notin \text{range}(A^\top)$) the risk row binds, so $\lambda = \sqrt{g^\top \Sigma^{-1} g}/(2\sigma_m)$ and $A^*_{\text{eq}} = \alpha^\top w^* = g^\top w^* = \sigma_m \sqrt{g^\top \Sigma^{-1} g}$ ($\nu^\top A w^* = 0$). Divided by the unconstrained $\sigma_m \sqrt{\alpha^\top \Sigma^{-1} \alpha}$, the retained fraction

$\sqrt{g^\top\Sigma^{-1}g/\alpha^\top\Sigma^{-1}\alpha}$ equals TC at w^*, and it is at most one by Cauchy–Schwarz in the Σ^{-1} metric ($g^\top\Sigma^{-1}g = g^\top\Sigma^{-1}\alpha$, as $A\Sigma^{-1}g = 0$), since g is the Σ^{-1}-orthogonal projection of α onto $\{v : A\Sigma^{-1}v = 0\}$ along $\mathrm{range}(A^\top)$.

A4 (alignment). Let B have full column rank. $B^\top D^{-1}\alpha_\perp = 0$ (the regression residual) gives $\Sigma D^{-1}\alpha_\perp = (D + BFB^\top)D^{-1}\alpha_\perp = \alpha_\perp$, so $\Sigma^{-1}\alpha_\perp = D^{-1}\alpha_\perp$ and $B^\top\Sigma^{-1}\alpha_\perp = 0$, with no assumption on F: under the risk budget alone $\alpha_\perp$'s optimum, proportional to $\Sigma^{-1}\alpha_\perp$, has zero factor exposure, and factor bounds $b_k > 0$ have zero shadow price. With $A = B^\top$, $g = \alpha_\perp$: the projection of A2 is unique, and $\alpha_\perp$ satisfies both $B^\top\Sigma^{-1}\alpha_\perp = 0$ and $\alpha - \alpha_\perp = \alpha_\parallel = B\gamma \in \mathrm{range}(B)$ (γ the regression coefficients). The cross term $\alpha_\perp^\top\Sigma^{-1}\alpha_\parallel$ vanishes, so $\alpha^\top\Sigma^{-1}\alpha = \alpha_\perp^\top D^{-1}\alpha_\perp + \alpha_\parallel^\top\Sigma^{-1}\alpha_\parallel$, and full factor neutrality retains $\sqrt{1-\varphi(\alpha)}$ of the alpha.

A5 (envelope). By Slater's condition (A1) strong duality holds. A^* as a function of the right-hand sides is the perturbation function of a convex program, hence concave, and nondecreasing in each bound because loosening it enlarges the feasible set [12]. Where the multipliers are unique (for example under LICQ), $dA^*/dr_k = \mu_k$, also across nondegenerate active-set changes, where μ_k has a kink but A^* does not; elsewhere only one-sided slopes exist, as at a cap tie $u = \theta\,\mathrm{ADV}_i/\mathrm{AUM}$. The chain rule gives the family dual of Sec. II-D.

A6 (perspective tightness). Over the same (w, z), restricted by the same linking rows, $\zeta \in (0, 1]$ gives $q \geq w^2/\zeta \geq w^2$ (and $w = 0$ at $\zeta = 0$), so with $\lambda^* \geq 0$ the perspective relaxation's risk term is at least big-M's and its value at most big-M's.

A7 (one-asset toy). For $\max\, aw - \lambda d w^2$ ($\lambda d > 0$) over $w \in \{0\} \cup [\ell, u]$, $w \leq b$, the value is 0 for $b < \ell$ and jumps at $b = \ell$ by $\max(0, a\ell - \lambda d\ell^2)$, positive iff $a > \lambda d\ell$. Below ℓ the true local slope is 0, and so is the fixed-integer dual. For $b < \ell$ the big-M relaxation's slope is $a - 2\lambda d b$ if $b < w^\circ = a/(2\lambda d)$ and 0 otherwise; the perspective relaxation's (q minimized at $z = w/\ell$, so $q = \ell w$) is $a - \lambda d\ell$ if $a > \lambda d\ell$ and 0 otherwise. For $\ell/2 < b < \ell$, $a - 2\lambda d b < a - \lambda d\ell$: where both are positive, big-M's slope is the closer to 0 although the perspective bound is the tighter. A tighter bound need not give a better slope.

A8 (integer value functions). The P-MI value $U^*(b)$ is the upper envelope, over support patterns, of concave pieces: nondecreasing by feasible-set inclusion, in general neither concave nor continuous. The relaxation dual is the slope of an upper-bounding concave function, the fixed-integer dual the slope of the active piece, and a re-solve gives a secant across pieces. With V the polished value, $\bar{U}$ Gurobi's final bound and $\bar{U}_{\text{eff}} = \max(\bar{U}, V) + 10^{-6}|V|$, $V \leq U^* \leq \bar{U}_{\text{eff}}$ at $b \pm \delta$ (δ absolute here, a percentage of b in the experiments) gives the certified interval

$$L = \frac{V_+ - \bar{U}_{\text{eff},-}}{2\delta} \leq \frac{U_+^* - U_-^*}{2\delta} \leq \frac{\bar{U}_{\text{eff},+} - V_-}{2\delta} = H, \quad (3)$$

with point slope $s = (V_+ - V_-)/(2\delta)$. "Certified" is conditional on V's weights being feasible (the independent checker passes them at 10^{-9}) and on Gurobi's bound being valid up to the slack, a margin, not a proof (Sec. IV-C reports a counter-case). A case is *certified binding* iff $L > 0$ (unlike *binds*, $e_p \geq 10^{-4}$, in the convex problems); $H < 0$ flags solver error. In I1 and I2 both formulations are solved at every point; V is the larger polished value, $\bar{U}$ the smaller bound, and the *certified gap* is $(\bar{U}_{\text{eff}} - V)/|V|$.

A9 (scaling). An absolute tolerance τ on a quantity of size m is a relative error τ/m; rescaling a row or the objective changes the effective tolerances of an equivalent problem. At an optimum with $\lambda^* > 0$ a held name has $q_i = w_i^2 \in [\ell^2, u_i^2]$ (q_i is minimized, not bounded), so Gurobi's absolute feasibility tolerance 10^{-6} is a relative error of up to $10^{-6}/\ell^2 = 0.16$ on the smallest cones, and at most $10^{-6}(u_i/\ell)^2 = 5.8 \times 10^{-4}$ with each cone in its name's bound. These scale a cone residual; they do not bound the objective error. They account for the incumbents' cone violation, with objective effect $\lambda^* \sum_i d_i \max(0, w_i^2/\zeta_i - q_i)$ reported relative to $|V|$, and for certified gaps above MIPGap: the default-scaling N2(a) solves at $s_{\text{obj}} = 10^4$ ended OPTIMAL (4 of 4), not at the WorkLimit, so their gap comes from the cone effect (Sec. IV-C). They do not account for the default-scaling root bound (Sec. IV-C).

D. Finite differences and noise

The perspective relaxation's slope at b is the central difference $\hat{s} = (R(b + \delta_r) - R(b - \delta_r))/(2\delta_r)$, $\delta_r = 10^{-3}\,b$, of its values R: the objective at Clarabel's (w, ζ) with each q_i set to w_i^2/ζ_i (0 where $w_i = 0$), so every cone holds exactly. A relaxation fails if its residuals exceed 10^{-8} or its duality gap exceeds $\varepsilon = 10^{-9}\max(|R|, |R_{\text{ref}}|)$, R_{ref} the value at the unperturbed bound. If every stencil value is within ε of exact, noise moves $\hat{s}$ by at most $\eta = \varepsilon/\delta_r$; the gap test does not prove this premise, so η is a noise scale. A relaxation's *gap* at b is $(R - V)/|V|$.

E. Validation

574 passing tests and logged checks with negative controls confirm the analysis: the QP and the Clarabel QCQP agree on A^* to 1.2×10^{-10} and on λ^* to 3.8×10^{-8} (A1); A2–A3 hold to 8.9×10^{-16}, the A4 split to 2.1×10^{-16}; on 202 one-asset toys both formulations match A7 at 8279 grid points to 1.4×10^{-17} (3 points within Gurobi's tolerance below ℓ excluded); P-MI with $K \geq N$, $\ell = 0$ reproduces P-RB's weights and A^* to 1.5×10^{-15} (largest relative error, in the weights). An independent feasibility checker passes every C1 and C2 solution and every polished V. Before I1 ran, the slope method passed a check fixed in advance on random toys (0 outside tolerance at 8399 points; 1 failed relaxation excluded) and on 39 pairs of objective scales (0 outside tolerance). At $10 \times s_{\text{obj}}$ the I1 perspective slopes agree within a tolerance fixed in advance at 48 of 48 cases (12 at 2009-03 as an acceptance check, the rest afterwards), moving at most 3.2×10^{-7} relative where certified binding ($n = 34$); none of the 240 I1 relaxations failed.

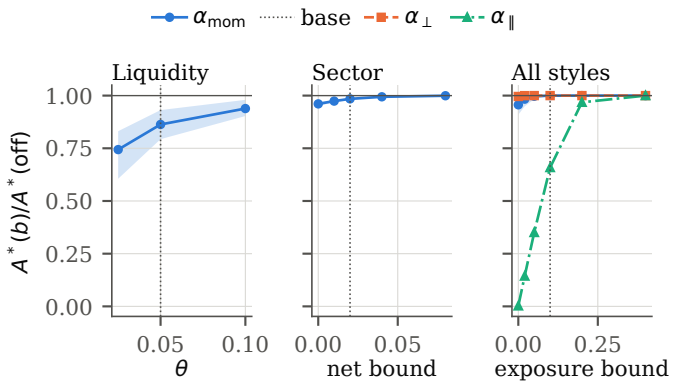


Fig. 1. Cost curves (C2): off = family removed, others at base (dotted: base); α_{mom} median and interquartile band over D12; all styles also $\alpha_{\perp}$, $\alpha_{\parallel}$.

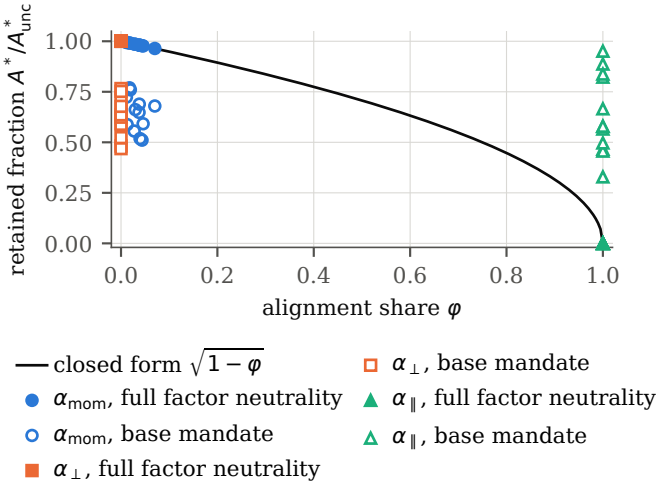


Fig. 2. Retained fraction against φ (C1), relative to the unconstrained, costless P-RB; filled: mandate (i), no costs; hollow: base.

IV. Results

A. Q1: Constraint Costs in the Convex Problem

Setup. C1 and C2 (Table II) at the D12 dates on all eligible names; the criterion is the elasticity e_p from exact duals, validated as derivatives below; statistics are over D12 for α_{mom} ($n = 12$) unless stated.

Which constraints bind. At base, A^* has median 30.5 bp/month (range 22.0–38.3), a median 0.65 (0.51–0.77) of the unconstrained, costless optimum. The risk budget binds in all but 4 of the 36 base solves, all of them $\alpha_{\parallel}$. The liquidity cap binds at all 12 dates (median elasticity 0.16), sector bounds at 12 (0.014), dollar neutrality at 10 (1.3×10^{-3}), style bounds at 4 and the position cap at 2; beta and gross bind at none. At the tightest grid points (Fig. 1) the pod keeps a median 0.744 of its alpha at $\theta = 2.5\%$ (range 0.52–0.95), 0.961 under exact sector neutrality, 0.956 with every style exposure at zero and 0.968 with the Mom exposure alone at zero; each other family keeps at least 0.994, the lowest being gross at $G = 1$.

What determines the cost. Factor bounds are cheap because α_{mom} is weakly aligned with the factors: φ is 0.011–0.070 over D12, and under idealized full factor neutrality the retained fraction is $\sqrt{1-\varphi}$ (A4), at least 0.964 (Fig. 2). Over the 72 idealized solves, numerical and closed-form A^* agree to 1.6×10^{-14} relative ($n = 48$; 3.6×10^{-17} absolute at the 24 $\alpha_{\parallel}$ solves where it is zero), and under mandate (i) ($n = 36$) the fractions match $\sqrt{1-\varphi}$ to 1.9×10^{-15}, or 7.5×10^{-8} for $\alpha_{\parallel}$, whose φ is within 5.7×10^{-15} of 1. $\alpha_{\perp}$ ($\varphi \leq 3.1 \times 10^{-33}$) loses nothing and $\alpha_{\parallel}$ ($\varphi = 1.0$) everything. At the base mandate the α_{mom} points lie well below the curve: the cost comes from the caps and the costs. Of the base portfolio's gross expected alpha $\alpha^{\top} w^*$, the pod keeps a median 0.80 (0.74–0.84) as A^*, pays 0.16 (0.13–0.21) in trading and 0.039 (0.032–0.052) in borrow.

Duals as derivatives. At every binding key and D12 date (115 checks over 18 keys) the dual matches the central difference within 1% (largest error 3.1×10^{-4}) at $\delta = 1\%$ of the bound, or at 0.1% at the 64 flagged stencils (the dual zero at an end, or forward and backward differences more than 1% of the central one apart); along all 312 C2 sweeps A^* is nondecreasing and concave in the bound (0 failures), as A5 predicts.

B. Q2: Sensitivity under Integrality

Setup. I1 (Table II) perturbs 4 bounds by δ (sector and dollar at 2%, all styles at 0.10, the Mom bound at 0.025): 48 cases, 36 at $\delta = 10\%$. The estimators are the big-M relaxation dual, the perspective relaxation slope and the fixed-integer dual; the baseline (Sec. II-E) is the convex reference. The criterion is $|\hat{\mu} - s|/|s|$ over certified-binding cases, and whether $\hat{\mu} \in [L, H]$.

Certification. No interval is violated (0 of 144 points) and the formulations' polished values tie at all 144. At $\delta = 10\%$, 23 of 36 cases are certified binding (sector 9, momentum 6, dollar 5, all styles 3); in the others every estimator is zero (the perspective slope to within 1.4×10^{-11}). The median relative width $(H - L)/s$ is 3.6×10^{-4} (1.2×10^{-4}–0.020, $n = 23$): lying in $[L, H]$ means matching the secant to that width.

Estimator errors. Table III gives the errors at $\delta = 10\%$ ($n = 23$). The perspective slope has the smallest median error, 0.013 (max 0.82), then the fixed-integer dual, 0.023 (max 0.82), and the big-M dual, 0.073 (max 4.2). Only 4 estimates, all fixed-integer duals, lie in $[L, H]$. The baseline row counts the 24 cases where the baseline's own constraint binds ($e_p \geq 10^{-4}$): the 23 certified-binding cases and 1 more. Its dual is within a median 3.7×10^{-4} (max 4.2×10^{-3}) of its own secant, so integrality, not the step, causes the errors. The errors depend on δ: at 2017-12 ($n = 11$ binding at each step) the medians at $\delta = 5\%$ and 10% are 0.014 and 0.023 (fixed-integer), 0.052 and 0.016 (perspective), 0.24 and 0.073 (big-M).

When the fixed-integer dual is exact. Each case is classed by comparing, at $b - \delta$, b and $b + \delta$, the signed support of V's incumbent and the fixed-integer QP's active set: P1, both unchanged; P2, support unchanged, active set changed; P3, support changed. Of the certified-binding cases, 3, 2 and 18

TABLE III
I1 ERRORS $|\hat{\mu} - s|/|s|$ AT $\delta = 10\%$, CERTIFIED-BINDING CASES, BY f.

estimator	f	n	median	max	in $[L, H]$	med. $(H - L)/s$
big-M dual	0.25	7	2.6×10^{-1}	4.2×10^{0}	0	6.2×10^{-4}
big-M dual	0.5	8	1.4×10^{-1}	1.2×10^{0}	0	3.2×10^{-4}
big-M dual	0.75	8	1.9×10^{-2}	3.0×10^{-1}	0	3.2×10^{-4}
perspective	0.25	7	4.5×10^{-2}	8.2×10^{-1}	0	6.2×10^{-4}
perspective	0.5	8	8.2×10^{-3}	1.0×10^{-1}	0	3.2×10^{-4}
perspective	0.75	8	8.4×10^{-3}	5.2×10^{-2}	0	3.2×10^{-4}
fixed-integer	0.25	7	4.8×10^{-2}	7.8×10^{-1}	1	6.2×10^{-4}
fixed-integer	0.5	8	6.0×10^{-2}	8.2×10^{-1}	0	3.2×10^{-4}
fixed-integer	0.75	8	5.2×10^{-3}	2.9×10^{-2}	3	3.2×10^{-4}
convex baseline	all	24	3.7×10^{-4}	4.2×10^{-3}	–	–

of 23 are P1, P2 and P3 at $\delta = 10\%$, and 4, 1 and 6 of 11 at $\delta = 5\%$. With fixed support and active set the QP solution is affine in b, so the value is quadratic and the central secant is the derivative at b. Over both steps, the fixed-integer error is at most 6.7×10^{-12} in P1 ($n = 7$), within the relative 10^{-6} predicted before the analysis ran, at most 2.9×10^{-4} in P2 ($n = 3$), and has median 0.056 in P3 ($n = 24$). The relaxation slopes fail differently: their error rises with the relaxation's gap at b (Spearman rank correlation 0.64 for the perspective, 0.50 for big-M, $n = 23$).

Value functions. I2 sweeps the sector bound, the I1 constraint with the largest median point slope over the 9 I1 instances at $\delta = 10\%$: 1.364×10^{-3} per unit of weight, against 1.316×10^{-3} per unit of loading for the Mom bound, a comparison across units; by median elasticity the Mom bound ranks first (0.027 against 0.020). The P-MI value lies below the baseline value and is not concave: 12 of 39 grid triples at 2017-12 and 9 of 39 at 2020-03 are certified nonconcave, $V(b^{(k-1)}) + V(b^{(k+1)}) > 2\bar{U}_{\text{eff}}(b^{(k)})$, and the support changes between 16 of 40 (2017-12) and 17 of 40 (2020-03) neighbouring grid points. At $0.5b$, b and $1.5b$ ($\delta = 10\%$, $n = 6$; Fig. 3) the median errors are 0.32 (big-M), 0.078 (perspective) and 0.059 (fixed-integer), none in its interval (relative width at most 9.9×10^{-4}).

C. Q3: Formulations and Numerics

Setup. N1 (Table II) solves the 12 instances per tightness level (loose: styles 0.4, beta 0.2; base; tight: 0.025, 0.0125) with the three arms at MIPGap 10^{-4} and WorkLimit W^*. The criteria are Work and the root gap after Gurobi's root cut loop, (root bound − best known value)/|best known value|, the best known value being the largest polished V over the instance's arms and presolve settings. Default-scaling results describe how Gurobi treats the unscaled cones (solver behaviour), not the formulation.

Root gaps and effort. At default presolve and base tightness the rescaled perspective has the smallest median root gap, 3.3×10^{-4} against big-M's 4.4×10^{-4} (big-M's is smaller at tight; Fig. 4), and the least median Work at every level (base 0.66 against 1.0); the default-scaling perspective is far behind (0.015; Work 1.2). No solve at default presolve stopped at the WorkLimit. The plain relaxations order as A6 predicts on every instance (0 violations; median margin 0.057 of $|V|$), with median plain gaps 0.059 (big-M) and 1.8×10^{-3} (perspective) at base ($n = 12$). Presolve, not the formulation, closes big-M's gap: at $N = 250$ ($n = 12$ per arm), presolve off raises big-M's median root gap from 6.0×10^{-4} to 0.057 (plain: 0.060), and 6 of 12 solves stop at the WorkLimit; the rescaled perspective moves only from 4.9×10^{-4} to 5.5×10^{-4} (median Work 0.766 to 2.28).

Two mechanisms. The default-scaling root bound is not the cone tolerance. N3 (Table II) repeats the root of both perspective arms at every N1 instance at default presolve, as root-only runs (*probes*); the default probe reproduces N1's root bound exactly at 36 of 36. Forcing outer approximation (MIQCPMethod 1; the default, −1, lets Gurobi choose) gives the same bound exactly at 29 of 36 (rescaled: 30), consistent with Gurobi choosing outer approximation at most instances; its logs are off, so the choice itself is not recorded. A QCP node relaxation (method 0, which solves each node's continuous quadratically constrained relaxation) lowers the median default-scaling root gap from 0.015 to 5.7×10^{-4}, below the plain perspective relaxation's 1.9×10^{-3} over the same 36 instances, as root cuts allow, whereas the default's root bound is looser than the plain relaxation itself, consistent with an outer approximation of the cones that stops before it converges. FeasibilityTol 10^{-9} leaves it at 0.015: the ratio of the two gaps has median 0.98, within a factor 2 at 31 of 36 instances. N3 was added after the other experiments had been analysed, to test on all 36 instances what the pilot run had found on a subset of them. The incumbent mechanism is the cone tolerance (A9): at default presolve ($n = 36$ per arm) the default-scaling incumbents violate their cones within tolerance with an objective effect of median 4.6×10^{-4} and max 6.4×10^{-3} of $|V|$, at most 1.9×10^{-11} rescaled. The rescaled arm changes both the rows' scale (divided by $\max_i u_i$) and the cones' scale, so N1 does not separate the two. Every arm's polished value lies within every other arm's $\bar{U}_{\text{eff}}$ (0 failures in 144 pairs).

Objective scaling. N2(a) repeats N1's base instances at $N = 250$ at $s_{\text{obj}} = 1$ and 10^4. At $s_{\text{obj}} = 1$ big-M stops at the WorkLimit in 2 of 4 instances (the others use 2.9–4.7 Work), and the rescaled perspective at 2017-12 with $K = 67$ reported a closed gap while its $\bar{U}_{\text{eff}}$ lay $9.1 \times 10^{-5}|V|$ below big-M's polished value, a false certificate. At 10^4 no incumbent exceeds another arm's bound (0, against 1), and the default-scaling arm's median certified gap is 1.2×10^{-3}, above MIPGap 10^{-4}: its cone effect reaches 4.8×10^{-3} of $|V|$, while the rescaled arm's median gap is 9.0×10^{-5}. In N2(b), P-RB at 2008-10, 2013-06, 2017-12, 2020-03, 2025-04, scaling the objective by 10^3 cuts Gurobi's barrier errors relative to the reference, worst over the dates and both risk-row normalizations ($n = 10$ each), from 3.1×10^{-5} to 7.2×10^{-7} in A^* and from 0.057 to 9.5×10^{-3} in the binding duals, while risk-row normalization (the risk cone divided by σ_m, to a unit right-hand side) leaves its A^* identical (10 of 10).

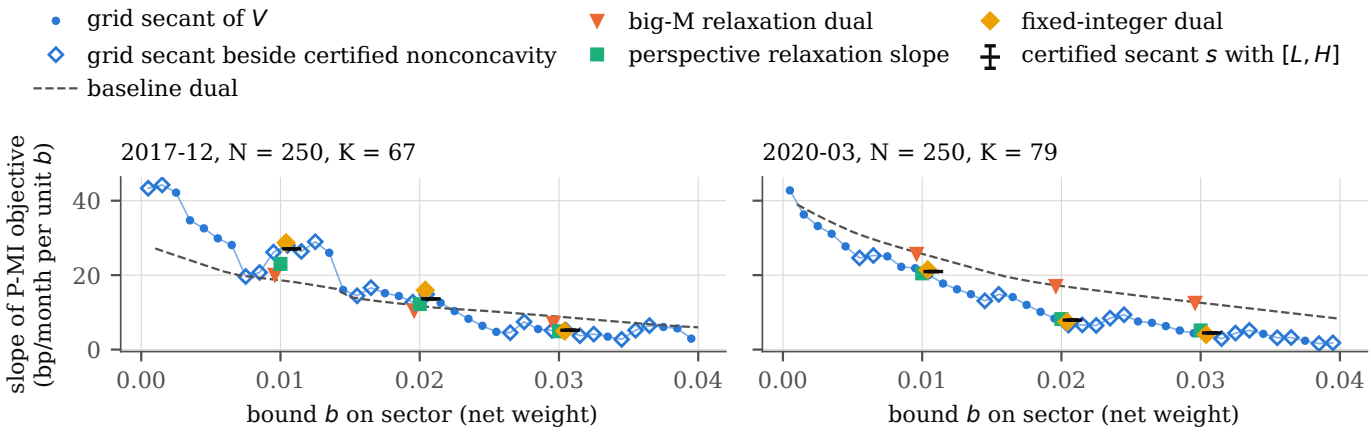


Fig. 3. I2 slopes on the sector bound (250 names; left 2017-12, right 2020-03), P-MI objective per unit of bound: neighbour secants of V (hollow: certified nonconcave), the baseline's dual, and the estimators beside s and $[L, H]$ at $0.5b$, b, $1.5b$ (offset).

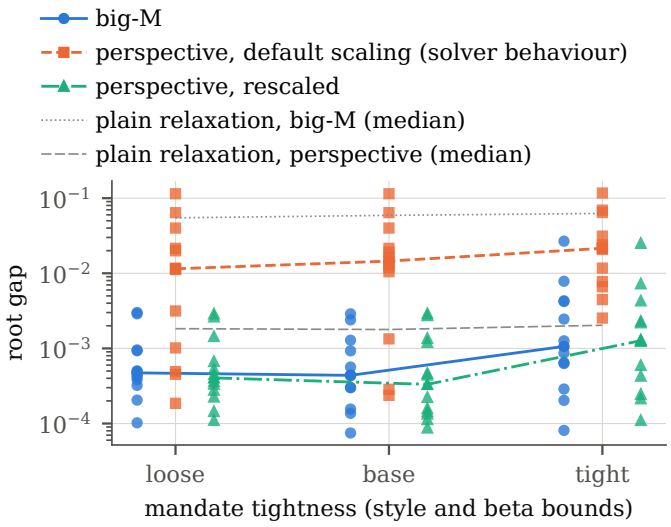


Fig. 4. N1 root gap against tightness, default presolve; $n = 12$ per cell; gray: median plain gaps.

V. Discussion

Fixing the risk budget rather than λ (Sec. II-D) is what makes Q1's rule costs comparable at equal risk. What the pod gives up comes mainly from the liquidity cap (median elasticity 0.16, against 0.014 for the sector bounds, which bind as often) and from trading and borrow, which take 0.16 and 0.039 of gross alpha. Factor neutrality is cheap because the momentum alpha is weakly aligned with the factors (φ from 0.011 to 0.070): idealized full factor neutrality keeps at least 0.964 of the unconstrained, costless optimum. The cost curves (Fig. 1) show the trade-off between alpha and exposure: relative to the book without each family, the pod keeps 0.744 at $\theta = 2.5\%$ and 0.961 under exact sector neutrality.

P-MI holds λ^* at its P-RB value for each date and universe, so Q2 measures marginal costs of the risk-penalized problem, not costs at equal risk, and its volatility drifts (median 2.87%/yr in I1, with 3 of 144 points above the budget). Holding the budget exactly would need the risk row as a constraint: with integers the Lagrangian relaxation of Sec. II-E can have a gap, since the optimal risk can jump as λ moves. The other modeling choices are assumptions (Sec. II, Table I): the start from cash, which charges trading cost on the whole book; the 252-day factor risk model and its residual floor, on which φ also depends; the momentum signal and its winsorization; the base mandate; P-MI's 6% cap and minimum position ℓ; and the cost rates. The disclosures below list their limitations.

As A8 sets out, a relaxation's slope is that of an upper bound, the fixed-integer dual that of the active piece, and a re-solve a secant across pieces. The certified intervals are narrow (median relative width 3.6×10^{-4}), so the errors are the estimators' own. With the support and active set fixed, the value is quadratic in the bound and the fixed-integer dual is exact (error at most 6.7×10^{-12}, $n = 7$); once the support changes, its median error is 0.056 ($n = 24$). Relaxation slopes err more as their relaxations loosen (Spearman 0.64 for the perspective, 0.50 for big-M, $n = 23$). At the one date with both steps (2017-12, $n = 11$ each), the fixed-integer dual leads at $\delta = 5\%$ (median 0.014 against 0.052) and the perspective slope at 10% (0.016 against 0.023), consistent with a longer secant crossing more pieces. So the fixed-integer dual can be trusted when the support is stable, but that shows only after re-solving; a price that drives a decision deserves a certified re-solve.

For Q3, strength trades against per-node cost: the perspective model's plain relaxation is far tighter than big-M's (median gaps 1.8×10^{-3} against 0.059 at base), and rescaled it needs fewer but costlier nodes (6.5 against 36.5) and less Work (0.66 against 1.0; medians over the 12 instances per arm and level). Presolve does most of big-M's root work: after presolve and root cuts the root gaps are close (3.3×10^{-4} against 4.4×10^{-4}), and at $N = 250$ turning presolve off raises big-M's from 6.0×10^{-4} to 0.057. A tighter mandate raises Work only modestly (to 0.77 rescaled and 1.1 big-M), though big-M's

root gap becomes the smaller (Fig. 4). Tolerance trades against runtime: the certified intervals widen with the solver's gap, which the Q2 runs hold to 10^{-6} with a WorkLimit of $2W^*$ (N1: 10^{-4}, W^*); with the objective unscaled, big-M stopped at the WorkLimit in 2 of 4 instances and even the rescaled arm gave a false certificate, its $\bar{U}_{\text{eff}}$ $9.1 \times 10^{-5}|V|$ below big-M's polished value. Most surprising, the default-scaling root bound is looser than its own plain relaxation (median gaps 0.015 against 1.9×10^{-3}, $n = 36$). Forcing outer approximation reproduces it at 29 instances, as it does the rescaled arm's small gap at 30, and QCP node relaxations cut it to 5.7×10^{-4}, consistent with outer approximation stopping early at default scaling; Gurobi's logs, or rescaling the cones without the rows, would test it.

An earlier version of this project, whose integer results were undermined by solver numerics, taught this one its fixed risk budget and scaling policy. As extensions, pricing exposures across the platform, a decomposition [13], would allocate exposure among pods by equalizing their shadow prices; model predictive control would rebalance from the current book, so trading costs follow turnover; an alpha uncertain within an ellipsoid gives a robust SOCP [12], which would show what neutrality buys, not only what it costs; and a Lagrangian bound on the cardinality row [13] would price the name limit, which has no derivative.

VI. Conclusion

At equal risk, the factor bounds are cheap for the momentum alpha because it is weakly aligned with the factors: idealized full factor neutrality keeps at least 0.964 of the unconstrained, costless optimum, and Mom at zero, the costliest single factor bound, keeps 0.968 of the alpha earned without it. The cost lies instead in the liquidity cap, trading and borrow. Under integrality, the fixed-integer dual is exact while the holdings and active set stay fixed within the step (error at most 6.7×10^{-12}) but not once the holdings change (median 0.056), and at one date the best cheap estimator changed with the step: a price that matters needs a certified re-solve. At default scaling, solver behaviour, not the formulation, sets the perspective model's root bound, through an outer approximation inferred to stop early, and its incumbents, through the cone tolerance; its tighter relaxation reaches the root only with rows and cones rescaled together, presolve brings big-M close, and the objective still needs scaling. Extensions (Sec. V) include pricing exposures across pods, multi-period rebalancing, a robust alpha and a Lagrangian bound on the name limit.

Disclosures and AI-Use Acknowledgment

The universe is the S&P 500 constituent list of 2026-10-16 applied to every past date, so survivorship and look-ahead enter it, and its GICS sectors are not point-in-time. Yahoo adjusted data come via yfinance, with errors for JCI and DHR masked; Sec. II-B's validity rules screen thin trading, and a multi-session return enters the regressions as a one-day return (1 in the panel). Re-standardizing after one pass of winsorization at ± 3 leaves $|z|$ up to 5.47 (2024-04) and above 3 at 240 of 240 month-ends and 1936 of 107592 name-dates. The problem is single-period from cash, the costs in Table I are assumptions, and the alpha is a simple input signal (IC 0.03), with no backtest or performance claim. Rules fixed in a pilot run before the experiments set s_{obj}, W^*, the K grids and the I1 bounds (Sec. III). A first, pre-registered version of this project was stopped when solver numerics undermined its integer results; this version, restarted as a course project, reuses its data, risk model and optimizer code but none of its results.

AI use. Claude (claude.ai chat) designed this version; proposed most design decisions, checks and design edits; drafted the agent's prompts and the decision notes; sketched derivations; drafted the proposal (reviewed by the author) and the Abstract, Sections I, V and VI and this section (approved by him); and proposed the edits to Sections II to IV that he chose. Claude Code, with result-checker, cold-reader and python-reviewer subagents, wrote and ran the code, tests and checks, built the figures, tables and macros, and drafted Sections II to IV. A separate AI chat without the project's context re-derived the Section II and III derivations before the author checked them by hand. The author chose the project, approved every recorded decision, and ran Claude (chat)'s scripts and every Claude Code session.

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